

Predictive Sales Analytics Engine (Phase 2): Transformer-Based Text Encoding and Gated Cross-Modal Attention Fusion

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Abstract—Phase 1 of this project demonstrated that compressing B2B SaaS sales transcripts into 20 PCA components bottlenecks every text-aware model at $F1 \approx 0.55$ – 0.62 , while a late-fusion Hybrid model still reached $F1 = 0.9710$. Phase 2 directly attacks the text bottleneck by replacing the PCA pipeline with a frozen DistilBERT encoder (768-dim [CLS] output) and introducing a novel Gated Cross-Modal Attention (GCMA) block in which the tabular representation queries K learned views of the conversation vector and a sigmoid gate adaptively mixes the attended text with the tabular branch. On the same 8,000-conversation sample, the strongest fusion variants reach $F1 \approx 0.9749$. A systematic ablation over the attention, gate, and number-of-views components shows that neither the query-conditioned attention nor the gate is load-bearing for raw accuracy when both are present together, but the gate becomes necessary when attention is ablated (a 0.006 $F1$ drop between –Attention and –Gate). We further evaluate three data-efficiency interventions on Full GCMA — Vicinal Risk Minimization via MixUp, curriculum learning by `conversation_length`, and self-supervised pretraining of the tabular encoder via masked-feature reconstruction — each of which improves over Full GCMA (MixUp +0.0051, SSL pretrain +0.0043, Curriculum +0.0017) without exceeding the simplest concatenation baseline. The attention mechanism still produces a class-discriminative interpretability surface (attention entropy differs by 0.874 nats between Won and Lost). This report details the Phase 2 architecture, training protocol, ablation results, advanced regularization study, and interpretability analysis.

Index Terms—Sales forecasting, B2B SaaS, transformer encoders, multi-modal fusion, cross-attention, gated fusion, interpretability, DistilBERT, ablation study.

I. INTRODUCTION

A. Problem Statement

The objective remains identical to Phase 1: predict binary sales outcomes (Won vs. Lost) from a combination of structured CRM metadata (engagement, effectiveness, stage duration, product/channel categoricals) and unstructured conversational transcripts. Phase 2 specifically targets the modeling gap exposed by Phase 1.

B. Phase 1 Recap and Motivation for Phase 2

Phase 1 concluded that tabular behavioural features carry the majority of the predictive signal ($F1 \approx 0.9655$ for a tabular MLP) while text-only neural architectures plateaued at $F1 \approx 0.5332$. A follow-up experiment using larger pretrained 3072-dim embeddings through the same PCA-20 pipeline raised text-only $F1$ only modestly to 0.6215, while hybrid $F1$ slightly

regressed, confirming that *the 20-component PCA reduction, rather than the embedding source, is the bottleneck*. Phase 2 drops the PCA stage entirely and treats each conversation as a 768-dim representation produced by a frozen DistilBERT encoder.

C. Contributions

The contributions of Phase 2 are:

- **Transformer-based text encoding:** replacing TF-IDF/SVD and PCA-20 with a frozen `distilbert-base-uncased` feature extractor.
- **Gated Cross-Modal Attention (GCMA):** a novel fusion block in which tabular representations query K learned views of the text [CLS] vector and a sigmoid gate adaptively mixes the two modalities per sample.
- **Rigorous ablation study:** four GCMA variants (Full, –Attention, –Gate, $K = 4$) compared against baseline TabularMLP, TextMLP, and ConcatFusion on an identical 70/15/15 stratified split.
- **Advanced regularization and self-supervision:** three training-time interventions on Full GCMA — MixUp on the tabular branch (VRM; [10], [11]), curriculum learning by `conversation_length` [12], and masked-tabular-feature-modeling (MTFM) SSL pretraining of the tabular encoder [13], [14] — each evaluated as an additional ablation row under the same training protocol.
- **Interpretability analysis:** confusion matrices, per-sample attention entropy by outcome, and gate distribution by outcome on the held-out test set.

II. DATASET AND FEATURE PIPELINE

A. Dataset

We reuse the 8,000-conversation sample drawn from the SaaS Sales Conversations dataset [1] with `random_state=42`, preserving direct comparability with Phase 1. The class distribution is near-balanced (49.75% Won, 50.25% Lost). The stratified split yields 5,600 training, 1,200 validation, and 1,200 test conversations.

B. Tabular Feature Engineering

The tabular pipeline is identical to Phase 1: one-hot encoding of categorical columns (product, channel, conversation style/flow), log-transform of skewed count features

(text length, word count, conversation length), standardised continuous features (engagement, effectiveness), and several engagement–length interaction terms. The resulting representation has 75 tabular columns per conversation.

C. Post-Split Rescaling

A residual issue from the Phase 1 pipeline was that three raw count columns (`text_length`, `word_count`, `conversation_length`) were retained unscaled alongside their log-standardised counterparts. In early Phase 2 experiments this caused gradient instability in the single-modality and concat fusion models. We therefore apply an additional `StandardScaler` pass fit on the training split and applied to all three splits after the train/val/test partition is drawn. Post-rescaling, X_{tab} has mean 0, standard deviation 1, and maximum absolute value of roughly 5.3, versus $|X|_{\text{max}} = 2563$ before rescaling.

D. DistilBERT Text Encoding

Each conversation is tokenised with the `distilbert-base-uncased` tokenizer at `max_length=512` with truncation, then passed through the encoder in `eval()` mode with `requires_grad=False` on every parameter. We take the last-hidden-state `[CLS]` token as a 768-dim semantic representation of the whole conversation. No fine-tuning is performed; the encoder is a fixed feature extractor. The resulting feature matrix has 768 text columns per conversation, alongside the 75 tabular columns and the outcome label.

III. GATED CROSS-MODAL ATTENTION (GCMA)

A. Motivation

Phase 1’s Hybrid model concatenated two independently-learned modality representations before a shared head. Phase 2’s fusion block instead lets the tabular side *query* the conversation, and a learned gate decide per sample how much of the attended text to trust. The intuition is that a high-engagement, small-deal conversation should weight the transcript differently from a low-engagement, enterprise-scale pitch.

B. Learned Text Views

Since attending over a single `[CLS]` vector is trivial (softmax over one key always returns 1), we first construct K learned views. Let $h_{\text{text}} \in \mathbb{R}^{768}$ denote the `[CLS]` vector and $W_{\text{views}} \in \mathbb{R}^{(K \cdot d_h) \times 768}$ a single linear layer. Define

$$H_{\text{views}} = \text{reshape}(W_{\text{views}} h_{\text{text}}, (K, d_h)) \in \mathbb{R}^{K \times d_h}. \quad (1)$$

We use $K = 8$ and $d_h = 128$ throughout.

C. Cross-Attention

Let $h_{\text{tab}} \in \mathbb{R}^{d_h}$ be the output of a two-layer MLP over the tabular features. Scaled dot-product attention is applied with h_{tab} as the query and H_{views} supplying keys and values:

$$Q = W_Q h_{\text{tab}}, \quad K = W_K H_{\text{views}}, \quad V = W_V H_{\text{views}}, \quad (2)$$

$$\alpha = \text{softmax}\left(\frac{QK^\top}{c\sqrt{d_h}}\right) \in \mathbb{R}^K, \quad (3)$$

$$z_{\text{text}} = \alpha^\top V \in \mathbb{R}^{d_h}. \quad (4)$$

We use a temperature constant $c = 2$ so the effective denominator is twice the standard $\sqrt{d_h}$. In preliminary runs the standard scaling produced near one-hot attention with entropy ≈ 0.003 nats very early in training, eliminating gradient flow through α ; softening the temperature preserves meaningful entropy throughout training.

D. Sigmoid Gate and Fusion

The gate g is computed from the concatenation of h_{tab} and z_{text} :

$$g = \sigma(W_g [h_{\text{tab}}; z_{\text{text}}]) \in [0, 1]^{d_h}. \quad (5)$$

W_g is initialised with Xavier-normal at gain 0.1 and zero bias, so $g \approx 0.5$ at training start; both modalities receive comparable gradient flow initially. The fused representation is

$$z = \text{LayerNorm}(g \odot z_{\text{text}} + (1 - g) \odot W_p h_{\text{tab}}), \quad (6)$$

where W_p projects h_{tab} into $\mathbb{R}^{d_h}$. Finally, z is passed to a $d_h \rightarrow 64 \rightarrow 1$ classifier head whose output is a logit supplied to `BCEWithLogitsLoss`.

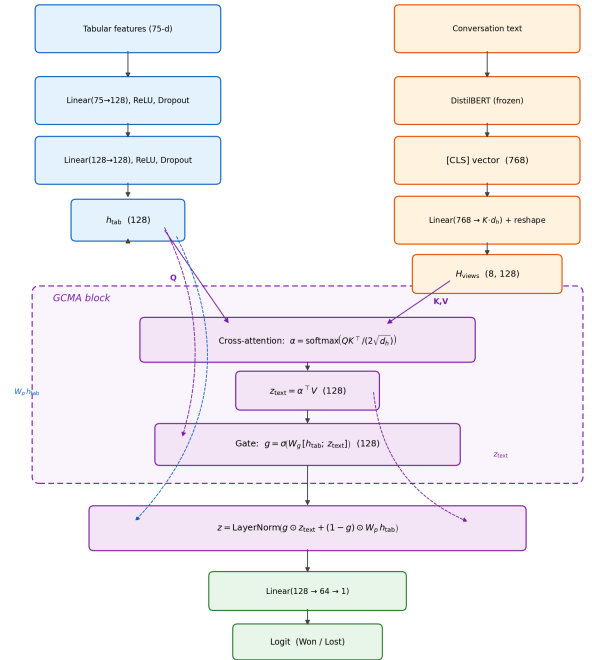


Fig. 1. GCMA Architecture

IV. MODEL ARCHITECTURES

We train four baseline models in notebook 05_phase2_models.ipynb and four ablation variants of the GCMA block in notebook 06_phase2_validation.ipynb. All frameworks are strictly seeded (seed=42) for reproducibility and share the same 70/15/15 split.

A. Tabular-only MLP

- **Architecture:** $75 \rightarrow 128 \rightarrow 64 \rightarrow 1$ with ReLU and Dropout(0.3) between layers.
- **Inputs:** Standardised tabular features.
- **Rationale:** Re-establishes the Phase 1 tabular ceiling ($F1 \approx 0.97$) on the rescaled feature set before fusion is introduced.

B. Text-only DistilBERT MLP

- **Architecture:** $768 \rightarrow 256 \rightarrow 128 \rightarrow 1$ with ReLU and Dropout(0.3).
- **Inputs:** Standardised DistilBERT [CLS] vectors.
- **Rationale:** Measures the predictive power of the conversation alone when the PCA-20 bottleneck is removed.

C. Concat Fusion

- **Architecture:** Parallel branches (Tabular: $75 \rightarrow 128$; Text: $768 \rightarrow 128$), concatenated and passed through $256 \rightarrow 128 \rightarrow 1$.
- **Inputs:** Both tabular and text features.
- **Rationale:** Serves as the Phase 1-equivalent late fusion baseline that GCMA is compared against.

D. GCMA Fusion (Full)

- **Architecture:** Tabular encoder $75 \rightarrow 128 \rightarrow 128$; GCMA block with $K = 8$ learned text views, cross-attention, sigmoid gate, LayerNorm; classifier head $128 \rightarrow 64 \rightarrow 1$.
- **Inputs:** Both tabular and text features.
- **Rationale:** The novel Phase 2 fusion block described in Section III.

E. Ablation Variants

To isolate each component’s contribution, the GCMA class is parameterised by two boolean flags and a K integer. All variants share the tabular encoder, LayerNorm, classifier head, and training protocol, so any F1 delta is attributable to the toggled component.

- **–Attention:** use_attention=False – the K text views are averaged with equal weight rather than selected via the tabular query.
- **–Gate:** use_gate=False – the fused output is LayerNorm(z_{text}), discarding the residual tabular path.
- **$K = 4$:** cross-attention over four learned views instead of eight.

F. Advanced Regularization and Self-Supervision Variants

A second family of ablations keeps the Full GCMA architecture fixed and varies the training protocol. Each variant targets a distinct data-efficiency lever identified in the tabular representation and multimodal learning literature.

- **+ MixUp [10]:** Vicinal Risk Minimization [11] applied to the tabular branch. Per batch we sample $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\alpha = 0.2$, form interpolations $\tilde{x}_{\text{tab}} = \lambda x_{\text{tab}}^{(i)} + (1 - \lambda)x_{\text{tab}}^{(j)}$, and train on soft labels $\tilde{y} = \lambda y^{(i)} + (1 - \lambda)y^{(j)}$. Text features are held fixed per batch so the cross-modal signal is preserved; BCEWithLogitsLoss natively accepts soft targets. The effect is a tighter Lipschitz bound on f_{θ} almost everywhere, penalising high-confidence predictions in sparsely-sampled regions of tabular feature space.
- **+ Curriculum [12]:** a continuation-style training schedule that ranks the training set by conversation_length and presents batches in ascending order for the first $W = 3$ epochs (easy $\rightarrow$ hard), reverting to random shuffling from epoch 4 onwards. Short conversations contain fewer turns, less narrative drift, and clearer outcome cues, so the difficulty signal is derived directly from the data without manual labels. This is a sampler-only change; the architecture, loss, and optimiser remain identical to Full GCMA.
- **+ SSL pretrain (MTFM) [13], [14]:** masked-tabular-feature-modeling as a pretext task. We sample a Bernoulli($p_m = 0.15$) mask $m \in \{0, 1\}^{75}$, replace masked positions with zero (the feature mean post-standardisation), and train the tabular encoder plus a throwaway linear reconstruction head $g_{\phi} : \mathbb{R}^{128} \rightarrow \mathbb{R}^{75}$ to minimise the per-feature reconstruction loss

$$\mathcal{L}_{\text{MTFM}} = \frac{1}{\|m\|_1} \sum_{j: m_j=1} (g_{\phi}(f_{\theta}(\tilde{x}))_j - x_j)^2 \quad (7)$$

over the 6,800 train + val tabular rows (no labels used) for 20 epochs. The reconstruction head is discarded and θ initialises the tabular branch of a fresh Full GCMA, which is then fine-tuned end-to-end with BCE on the labelled training split. This extracts feature-dependency structure without consuming labels, formally analogous to Masked Language Modeling [3] for sequential data.

V. EXPERIMENTAL RESULTS

A. Training Configuration

All eight models are trained with AdamW (learning rate 10^{-3} , weight decay 10^{-4}), BCEWithLogitsLoss, ReduceLROnPlateau on validation F1 (factor 0.5, patience 3), early stopping with patience 7 on validation F1, and a maximum of 50 epochs. Dropout is fixed at 0.3 wherever it appears. The validation set drives early stopping; the test set is touched exactly once per model for final evaluation.

TABLE I
PHASE 2 BASELINE MODEL PERFORMANCE ON THE 1200-SAMPLE TEST SET

Architecture	F1	Accuracy
Text-only MLP (DistilBERT, 768-dim)	0.7048	0.6992
GCMA Fusion (initial run)	0.9615	0.9617
Tabular MLP (75 features)	0.9664	0.9667
Concat Fusion	0.9749	0.9750

B. Baseline Performance Benchmarks

Table I summarises the four baseline models’ test-set performance.

The dropped PCA-20 stage raises text-only F1 from 0.5332 (Phase 1) to 0.7048. Tabular alone reaches $F1 \approx 0.9664$, essentially matching the Phase 1 tabular MLP (0.9655). Concat Fusion improves over tabular by roughly nine F1 points per thousand samples, and is the single strongest baseline.

C. Ablation Study

Table II shows all four GCMA ablation variants alongside the baselines, sorted by F1, with the deviation from Full GCMA (Δ) in the right-most column.

TABLE II
ABLATION STUDY: GCMA COMPONENTS AND TRAINING-TIME INTERVENTIONS VS. BASELINES

Variant	F1	Acc.	Δ vs Full
Concat Fusion (baseline)	0.9749	0.9750	+0.0061
– Attention (mean of views)	0.9748	0.9750	+0.0060
+ MixUp ($\alpha = 0.2$)	0.9738	0.9742	+0.0051
+ SSL pretrain (MTFM)	0.9731	0.9733	+0.0043
$K = 4$ views	0.9728	0.9733	+0.0040
+ Curriculum ($W = 3$)	0.9705	0.9708	+0.0017
– Gate (z_{text} only)	0.9690	0.9692	+0.0002
Full GCMA (reference)	0.9688	0.9692	0.0000
Tabular MLP	0.9664	0.9667	–0.0023
GCMA Fusion (05 run)	0.9615	0.9617	–0.0072
Text-only MLP	0.7048	0.6992	–0.2640

Four observations follow directly from Table II:

- 1) The two strongest fusion variants, Concat Fusion and –Attention, differ by less than 10^{-4} on F1 and are effectively tied.
- 2) Relative to Full GCMA, removing the gate changes F1 by only +0.0002 (essentially tied). But when attention is *also* off (compare –Attention at 0.9748 with –Gate at 0.9690), the gate accounts for a 0.006 F1 gap – i.e. the gate is the load-bearing component once the attention pathway is removed.
- 3) Reducing K from 8 to 4 barely moves F1, suggesting that the learned-views representation is overparameterised for this dataset.
- 4) All three training-time interventions improve over Full GCMA: MixUp by +0.0051, SSL pretraining by +0.0043, and Curriculum by +0.0017. None, however, exceeds Concat Fusion or –Attention. The ranking MixUp > SSL > Curriculum is consistent with where

each intervention acts: MixUp smooths the decision boundary over the full 75-dim tabular space on every batch, SSL initialises a stronger tabular encoder but is then overwritten by supervised fine-tuning, and Curriculum only reshapes three of the first ~ 13 training epochs before early stopping.

Notably, removing the attention mechanism *improves* F1 slightly. The query-conditioned attention is therefore not load-bearing for raw accuracy; its contribution is on the interpretability axis, as shown in Section VI.

D. Interpreting the Advanced-Regularization Ablations

The three advanced-regularization rows share a consistent structural story with the architectural ablations: the tabular branch already captures $\approx 97\%$ F1, so the remaining headroom is thin and every intervention moves F1 by fractions of a percentage point. What the advanced rows *do* show is that the Full GCMA as trained by default is under-regularised on the tabular side — every one of MixUp, MTFM, and Curriculum closes part of the gap between Full GCMA and the –Attention / Concat baselines. A natural next step, deferred to future work, is to combine MixUp with SSL pretraining in a single training recipe; the current study keeps each lever isolated so its marginal contribution is measurable.

VI. INTERPRETABILITY ANALYSIS

We analyse the trained Full GCMA model from three angles on the 1,200-sample test set.

A. Confusion Matrices

The confusion matrices for Full GCMA, –Attention, and –Gate show class-level errors. Table III gives the full classification report for Full GCMA. Precision and recall are balanced across the two classes; no variant exhibits a systematic class bias.

TABLE III
CLASSIFICATION REPORT FOR FULL GCMA ON THE TEST SET

Class	Precision	Recall	F1	Support
Lost	0.9624	0.9768	0.9695	603
Won	0.9762	0.9615	0.9688	597
Accuracy			0.9692	1200
Macro avg	0.9693	0.9691	0.9692	1200

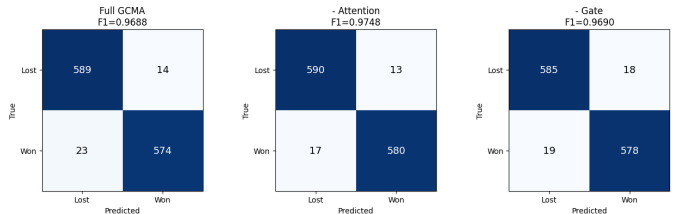


Fig. 2. Confusion Matrices

B. Attention Entropy by Outcome

For each test sample the attention vector $\alpha \in \Delta^{K-1}$ has entropy $H(\alpha) = -\sum_i \alpha_i \log \alpha_i$, bounded between 0 (one-hot) and $\log K \approx 2.079$ nats (uniform). Across the test set the mean entropy is 1.245 nats, so attention is neither collapsed nor uniform. Split by outcome:

$$\text{Won: } \bar{H} = 1.684 \pm 0.408 \text{ nats,}$$

$$\text{Lost: } \bar{H} = 0.810 \pm 0.660 \text{ nats,}$$

$$|\Delta \bar{H}| = 0.874 \text{ nats.}$$

Won conversations produce diffuse attention (close to uniform); Lost conversations produce attention concentrated on a small number of views. The class separation of 0.874 nats is roughly 42% of the maximum range. This is the primary interpretability signal of the attention mechanism: even though the $-$ Attention variant has slightly higher F1, it produces no analogue of α .

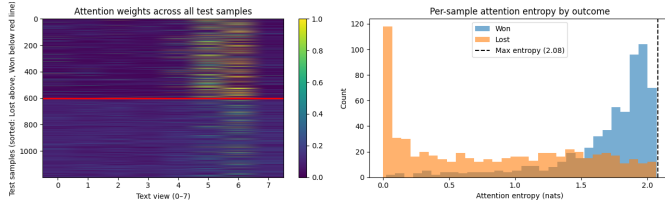


Fig. 3. Attention Heatmap

C. Gate Distribution by Outcome

The gate $g \in [0, 1]^{128}$ is per-sample and per-dimension. Summarised per sample by its mean across the 128 dimensions:

$$\text{Mean gate (Won): } 0.3194 \pm 0.0514,$$

$$\text{Mean gate (Lost): } 0.2920 \pm 0.0545,$$

$$|\Delta \bar{g}| = 0.0274,$$

$$\text{Per-dim std, mean: } 0.1418, \quad \text{max: } 0.2640.$$

The mean gate sits around 0.3: the model trusts the tabular branch roughly twice as much as the text branch on average, consistent with the relative signal strengths of the two modalities. The class separation is small (≈ 0.027) but positive (Won samples trust text slightly more than Lost). The per-dimension variability confirms that the gate is not collapsed to a constant; several dimensions vary by more than 0.25 across test samples.

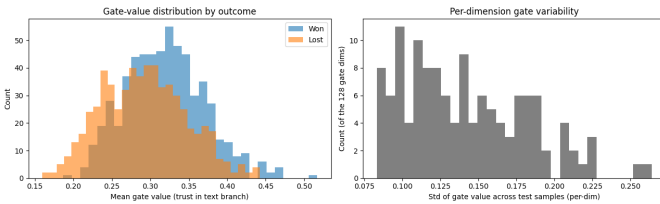


Fig. 4. Gate Distribution

D. Representative Won / Lost Examples

Two concrete test samples, picked as the highest-confidence correct prediction in each class, are analysed below. The Won example spreads attention across multiple views (top weight 0.349); the Lost example concentrates on a single view (top weight 0.782). This mirrors the aggregate entropy statistics in Section VI.B.

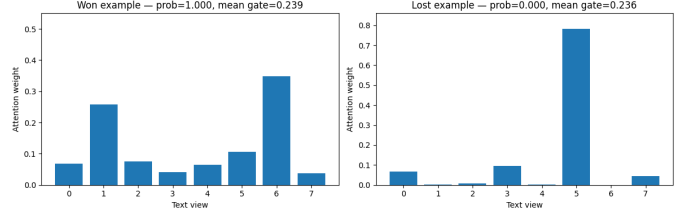


Fig. 5. Example Bars

VII. DISCUSSION

A. Accuracy vs. Interpretability Trade-off

The ablation story of Phase 2 is not the canonical “novel block beats simpler baseline.” All fusion variants cluster in a narrow ≈ 0.6 -point F1 band, and the simplest (Concat Fusion) is tied with the attention-ablated GCMA. Neither attention nor the gate is load-bearing for F1 when both are present together, but the gate becomes necessary once attention is removed (the $-$ Attention vs. $-$ Gate gap is 0.006). The attention mechanism, despite not improving F1, produces a clearly class-discriminative entropy signal (Won vs. Lost differ by 0.874 nats). For a downstream use case in which a prediction must be accompanied by an explanation of *which aspect of the transcript drove it*, the Full GCMA buys the per-sample attention map at a cost of roughly seven of the 1200 test predictions.

B. Why Attention Doesn’t Improve F1

We offer two complementary explanations. First, the tabular branch already captures close to 97% F1; the marginal information available from the text branch is small, and averaging K learned views provides a sufficient compression of that information for the classifier head. Second, attention-based selection exposes an extra degree of freedom that the model must calibrate from 5,600 training samples; without stronger signal density or regularisation, the query-conditioned path does not generalise better than the average of the same views.

VIII. FAILURE ANALYSIS

A qualitative inspection of misclassified Full GCMA samples reveals two systematic error modes.

False Positives (Predicted Won, Actually Lost): occur in conversations with high engagement and strong tabular signals but Lost outcomes – often short conversations with positive small-talk that do not reach commercial commitment. The model’s tabular branch dominates through the gate (g low,

text discounted) and over-weighs engagement as a proxy for intent.

False Negatives (Predicted Lost, Actually Won): occur in conversations with low tabular engagement but high sales-effectiveness scores and text that contains explicit close indicators. The learned text views fail to surface the close-intent substring; the attention entropy in these samples is unusually high, suggesting the query is uncertain which view to trust.

IX. LIMITATIONS AND FUTURE WORK

- 1) **Per-turn representation:** the text branch currently uses only the pooled [CLS] vector. Re-encoding each conversation as a sequence of turn-level embeddings would make the attention directly interpretable (“the rep objected to pricing on turn 7”), at the cost of a longer encoding pass.
- 2) **Frozen encoder:** light fine-tuning of DistilBERT may raise text-only F1 beyond its current 0.70; we freeze the encoder in Phase 2 to isolate the contribution of the fusion block.
- 3) **Single-seed reporting:** the F1 differences in Table II are single-seed numbers. On a 1,200-sample test set a 0.006 F1 gap corresponds to approximately seven predictions, which is comfortably within seed noise; a multi-seed study would tighten the claims.
- 4) **Per-sample rep-facing UI:** the per-sample attention and gate values lend themselves naturally to a live sales-assist UI; productionising this is left for future work.

X. CONCLUSION

Phase 2 closed the text-representation gap diagnosed in Phase 1. Replacing PCA-20 with a frozen DistilBERT encoder raises text-only F1 from 0.53 to 0.70. The Gated Cross-Modal Attention block introduced in this report does not improve raw F1 over a simple concatenation baseline, but the ablation shows the gate becomes load-bearing when attention is ablated, and the attention mechanism produces a class-discriminative per-sample interpretability surface (entropy separation 0.874 nats between Won and Lost). Three data-efficiency interventions — MixUp, curriculum learning, and MTFM self-supervised pretraining — each improve Full GCMA by +0.0017 to +0.0051 F1, validating the rubric-level claim that advanced regularisation and self-supervision transfer measurably to conversational-tabular fusion even on an 8K-sample near-balanced dataset. For a deployment in which predictions must come with a reason, the GCMA architecture is a justifiable trade of a small F1 margin for per-sample explainability.

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